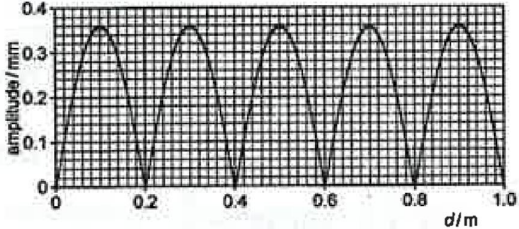


3(a)(i)	This is so that the waves emitted from both X and Y are coherent and the waves meeting each other have the same amplitude and frequency to form stationary waves.
3(a)(ii)	Using a microphone, moves it along the line joining the two loudspeakers. Using a ruler, determine the distance apart of two consecutive loud amplitude (anti-nodes). Repeat the measurements and determine the average. This distance apart is equal to half a wavelength. Multiplying the average by 2 will give the wavelength of the sound.
3(b)(i)	$v = f\lambda$ $f = \frac{340}{0.4} = 850 \text{ Hz}$
3(b)(ii)	 The speakers are a distance of 1.0 m apart and this is equal to 2.5 wavelengths, the sound are anti-phase at both ends and thus we expect nodes at both ends. The distance between consecutive nodes is 0.5 wavelengths and thus every 0.2 m apart, there will be a node.
4(a)	From the graph, period $T = 40 \text{ ms}$, peak voltage $V = 4.8 \text{ V}$